



Compositional Quality and Possible Gastrointestinal Performance of Marketed Probiotic Supplements

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Accepted: 11 February 2022 / Published online: 23 February 2022

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Abstract

The local pharmacies and shops are brimming with various probiotic products that herald a range of health benefits. The poor quality of probiotic products in both dosage and species is symptomatic of this multi-billion-dollar market making it difficult for consumers to single out reliable ones. This study aims to fill the potential gap in the labeling accuracy of probiotic products intended for human consumption. We describe a combinatorial approach using classical culture-dependent technique to quantify and molecular techniques (16 s rRNA gene sequencing, multilocus sequence, and ribotyping) for strain recognition of the microbial contents. The full gamut of probiotic characteristics including acid, bile and lysozyme tolerances, adhesiveness, anti-pathogenicity, and degree of safeness were performed. Their capacity to endure gastro-intestinal (GIT) stresses and select drugs was assessed in vitro. Our results forced us to declare that the local probiotic market is essentially unregulated. Almost none of the probiotic products tested met the label claim. Some (11%) have no viable cells, and a quarter (27%) showing significant inter-batch variation. A lower microbial count was typical with undesirables constituting a quarter of the total (~27%). Half of the products contained antibiotic-resistant strains; the unregulated use of these probiotics carries the risk of spreading antibiotic resistance to gut pathobionts. Poor tolerance to gut conditions and mediocre functionalism make the case worse. The current regulatory systems do not take this discrepancy into account. We recommend an evidence-based regular market surveillance of marketed probiotics to ensure the authenticity of the claims and product effectiveness.

Keywords Regulation · Commercial probiotics · Strain-typing · Antibiotic resistance · Gastrointestinal tract

Introduction

The positive modulation of complex microbial communities with the beneficial microbes is an ever-increasing popular option [1] which has given significant impetus to probiotic development [2, 3]: the probiotic dietary supplements being the lead eubiosis restorers in healthcare [4]. Since 2001, the probiotic sector has seen substantial growth in scientific research and commerce [5]. The global probiotic industry has an approximate market worth of \$15 billion with a 7% per annum growth rate [6] and predicted sales of billions \$91.1 by 2026 [7]. The probiotic trade in Asia-Pacific has outpaced that of Europe (EU), mainly because of the

relatively lax health claim regulations than in the EU [8]. Oral administration of probiotics still stands out among the various modulation strategies available and is significantly more effective and safer than fecal microbial transplantation (FMT) [9]. Probiotics go beyond digestive health and include moderating immunity, metabolic and respiratory and cardiac health, and even individual mood and temperament [10]. The haste for discovering novel microbes with potential therapeutic benefits has often resulted in distortions and misinterpretation of the probiotic concept, necessitating stricter adherence to the scientifically correct definition of probiotic [5]. The currently globally recognized definition of probiotics [11] distinguishes the probiotic from commensals and includes the non-human origin. The core of the definition is the microbial origin, viability, and eventual benefit [11]. A corollary of this is that their therapeutic effect depends on the dose taken, the viability of the bacteria at the site of action, and their level of metabolic activity [12].

Regulatory scrutiny of probiotics is critical to keeping consumers interested in their benefits. Poor reproducibility in the quality and performance of the end product can

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affect its marketability, even if it has good yield and stability [13]. Several studies have commented on the quality of the probiotics marketed, which is inconsistent with their claims [14–16]. In addition, the reports on infant probiotic supplements have been relatively worth considering [17–19], and the need for stricter quality control on probiotic infant formulas has been highlighted before [19, 20].

This study aims to fill the potential gap in the labeling accuracy of probiotic products intended for human consumption. Therefore, we have researched Pakistan's local probiotic market, focusing on dietary supplements and infant formula, to see if they live up to their labels in terms of quantity and quality. We also characterize the commercial strains in terms of their probiotic, functional, and safety properties.

Material and Method

Commercial Probiotic

Fifty-five samples purportedly containing live or inactivated probiotic strains were collected from retail outlets in three major cities in Pakistan, including 14 infant formula, 39 local pharmaceutical preparations, and two foreign ones (Table 1). These were replicates of 18 distinct, over the counter (OTC) formulations or brands which were most popular and widely available in the country. The storage conditions were retained after the purchase. The labels were examined for product content such as certain strains, probiotic dosages at the end of shelf life, and claimed health benefits. All formulations were purchased in person from pharmacies and analyzed before their stated expiration date.

Microbial Reference Strains

Microbial reference strains from international culture collections were used for media quality check and experimental purposes (Supplementary Table 1).

Analytical Procedures

A polyphasic approach was chosen for formulation analysis, comprising the following: (i) putative differential enumeration of labeled probiotic strains; (ii) detection of foreign bacteria, fungi, or coliforms; (iii) phenotypic and genotypic (16S rRNA-based) identification of the bacterial isolates obtained; (iv) strain-level variation between strains of the same species isolated from the same/different products by ribotyping and multilocus strain typing; (v) probiotic characterization and safety through a range of in vitro assays; (v) survival of probiotic strains under statically simulated GIT transit by compartmentalizing in vitro digestion models; (vi)

effect of non-antibiotic drugs on the survival of probiotic strains under simulated conditions.

Microbial Enumeration and Identification

The unit dose of the probiotic formulations was dissolved in sterile phosphate-buffered saline (PBS) immediately before analysis. All suspensions were serially diluted in PBS and plated out on various media to selectively count the strains contained. Approximately 100 μ L aliquots of the formulations were placed on De Man, Rogosa & Sharpe medium (Merck, Germany) supplemented with 0.05% L-cysteine hydrochloride monohydrate (MRS_c) for the isolation of *Lactobacillus* species. Transgalactosylated oligosaccharide agar supplemented with lithium mupirocin (TOS_{MUP}) for bifidobacterial species; Reinforced Clostridial Medium (RCM) selective for clostridial species; Kenner fecal streptococcal (KFS) agar supplemented with 1% (2, 3, 5-Triphenyl Tetrazolium Chloride) (KFS_{TTC}) for *Enterococcus* spp.; and M17 medium with lactose (M17_{Lac}) for *Streptococcus* spp. were also used. The MRS_c, TOS_{MUP}, and RCM plates were incubated at 37 °C for 48 h in an anaerobic atmosphere (H₂ 5%; CO₂ 5%; N₂ 90%) maintained in an anaerobic chamber (Bactron300-2, Shel Lab, USA). The KFS_{TTC} and M17_{Lac} were incubated aerobically at 37 °C. A possible bacterial, fungal, and coliform contamination of the products was checked by aerobic plating on Nutrient agar (NA, Himedia, India), Sabouraud Dextrose Agar (SDA), and MacConkey agar (MK, Neogen, UK), respectively. An automated spiral plater (Autoplater[®], Advanced Instruments Inc., USA) was used for aerobic plating. Plating was performed in triplicate. The number of CFU was determined using an automated colony counter (Qcount[™], Advanced Instruments, USA), and the data was saved as a user-defined file (Qcount software[®]); typical colonies were subjected to identification based on colony characteristics and biochemical properties and frozen in 20% (v/v) glycerol. Growth curves for determining growth kinetic parameters; specific growth rates (μ) and doubling time (Td) were calculated [21].

Bacterial cells were lysed to obtain DNA using a commercial kit (Favorgen, Taiwan), quantified, and qualified by NanoDrop and agarose gel electrophoresis, respectively. The PCR amplification of the V3-V4 regions of the 16 s rRNA gene was carried out using the primer pairs 341F and 806R (Supplementary Table 2) [22]. Purified PCR products (gel extraction kit, Thermo Fisher Scientific, USA) were subjected to nucleotide sequencing and subsequently deposited in GenBank, and their accession numbers were obtained. Phylogenetic affiliations were mapped out by aligning the 16S rRNA gene sequences with closely related BLAST reference sequences (Fig. 1), and their phylogenetic relatedness mapped out using the maximum likelihood tree (MLT) in MEGA version X [23] (Fig. 1).

Table 1 A comparison of claimed and actual total viable bacterial count (cfu/dose) and molecular identification in probiotic supplements

Product code	Claimed	City	Actual anaerobic count		Strain code	Identification	Actual aerobic count				NPL#	Identification			
			*Strain	cfu/dose			MRS _c	TOS _{MUP}	RCM	KF S _{TTC}			M17 _{Lac}	NA	SDA
PS1	<i>L. fermentum</i> Lc40	A	3 billion CFU/capsule	-	-	285	<i>L. fermentum</i>	-	-	-	0	0	0	NA	
		B		-	-	283		-	-	-	0	0	0		
		C		-	-	-		-	-	-	0	0	0		
IPS2	<i>L. rhamnosus</i> GG-125	A _{ref}	≥5×10 ⁹ CFU	-	-	282		-	-	-	0	0	0		
		A	(approx.)	-	-	730	<i>L. rhamnosus</i>	-	-	-	1.6×10 ⁷	0	0	817	
		B	sachet (2 g)	-	-	908		-	-	-	3.6×10 ⁶	0	0	828	
		C		-	-	-		-	-	-	0	0	0	-	
		A _{ref}		-	-	909	NA	-	-	-	5.4×10 ⁸	0	0	829	
PS3	DDS-1 (<i>Lact. acidophilus</i>)	A	37.5mg contains 0.375 billion viable cells	-	-	-		-	-	-	0	0	0	NA	
		B		-	-	-		-	-	-	-	-	-		
		C		-	-	-		-	-	-	-	-	-		
		A _{ref}		-	-	-		-	-	-	-	-	-		
		A	100 million live active cells per serving (5 drops)	-	-	-	-	<i>L. reuteri</i>	-	-	-	0	0	0	NA
PS4	<i>L. reuteri</i> Protectis	B		-	-	-		-	-	-	0	0	0		
		C		-	-	-	-		-	-	0	0	0		
		A _{ref}		-	-	-	729		-	-	0	0	0		
		A	Minimum 100 million live <i>L. reuteri</i> Protectis/dose/5drops	-	-	-	-		-	-	-	-	-		
		B		-	-	-	-	<i>L. reuteri</i>	-	-	-	0	0	0	NA
PS5	<i>L. reuteri</i> Protectis™ DSM 17938	C		-	-	-	-		-	-	-	-	-		
		A _{ref}		-	-	-	-		-	-	-	-	-		
		A		-	-	-	-		-	-	-	-	-		
		B	1.1×10 ⁸	-	-	-	338		-	-	0	0	0		
		C		-	-	-	-		-	-	-	-	-		
PS6 _{IF}	<i>L. Protectus</i> (<i>L. reuteri</i>)	A _{ref}		-	-	-	-		-	-	-	-	-		
		A	1.6×10 ⁷	-	-	-	284, 357	<i>L. caviae</i>	-	-	-	0	0	0	NA
		B	5.7×10 ⁶	-	-	-	366	<i>L. reuteri</i>	-	-	-	0	0	0	
		C	5.1×10 ⁶	-	-	-	364		-	-	0	0	0		
		A _d	5.8×10 ⁶	-	-	-	544		-	-	0	0	0		
PS7 _{IF}	<i>L. Comfortis</i>	A	8.8×10 ⁵	-	-	-		-	-	-	0	0	0	NA	
		B	7.6×10 ⁶	-	-	-	362	<i>L. reuteri</i>	-	-	-	0	0	0	
		C	4.8×10 ⁶	-	-	-	358		-	-	0	0	0		
		A _d	1.2×10 ⁶	-	-	-	360		-	-	0	0	0		
		A		-	-	-	541		-	-	0	0	0		
PS8	<i>Bif. subsp. lactis</i> (BB-12)	A	1 sachet contains not less than 1 × 10 ⁹ CFU	-	-	-		-	-	-	0	0	0	NA	
		A		-	-	-	355	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	-	0	0	0	
		B		-	-	-	-		-	-	-	-	-		
		C		-	-	-	356		-	-	0	0	0		
		A _{ref}		-	-	-	-		-	-	-	-	-		

Table 1 (continued)

Product code	Claimed	City	Actual anaerobic count			Strain code	Identification	Actual aerobic count				NPL#	Identification
			MRS _c	TOS _{MUP}	RCM			KF S _{TTC}	M17 _{lac}	NA	SDA	MK	
PS9 _{IF}	<i>Bifidus</i> B _L	A	-	1.0×10 ⁷	-	369	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	0	0	0	<i>Staph. epidermidis</i> <i>B. singularis</i> NA
		B	-	8.7×10 ⁶	-	601		-	-	6.7×10 ³	0	0	
		C	-	7.2×10 ⁶	-	602		-	-	4×10 ³	0	0	
		A _d	-	6.9×10 ⁶	-	370		-	-	0	0	0	
PS10 _{IF}	<i>Bifidus</i> B _L	A	-	7.2×10 ⁶	-	604	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	0	0	0	NA
		B	-	-	-	-		-	-	0	0	0	
		C	-	-	-	-		-	-	0	0	0	
		A _d	-	2.0×10 ⁷	-	368		-	-	0	0	0	
PS11	<i>Ent. faecium</i> SF68	A	-	-	-	-	-	-	-	-	-	-	<i>Ent. hirae</i>
		B	-	-	-	-		1.2×10 ⁸	1.3×10 ⁸	0	0	0	
		C	-	-	-	-		1.1×10 ⁹	1.2×10 ⁹	0	0	0	
		A _{ref}	-	-	-	-		9.1×10 ⁸	1.3×10 ⁹	0	0	0	
PS12	<i>L. rhamnosus</i> GR-1 and <i>L. reuteri</i> RC-14	A	0	-	-	-		-	-	TFTC	0	0	<i>S. asenjonii</i> <i>L. rhamnosus</i>
		B	9.8×10 ⁸	-	-	983,	<i>L. rhamnosus</i>	-	-	3.9×10 ⁸	0	0	
		C	-	-	-	984	<i>L. reuteri</i>	-	-	-	-	-	
		A _{ref}	-	-	-	-		-	-	-	-	-	
PS13	<i>Bifidobacterium</i> , BB-12 <i>L. rhamnosus</i> GG, LGG	A	4.1×10 ⁹	1.5×10 ¹⁰	-	741, 744	<i>L. rhamnosus</i>	-	-	0	0	0	NA
		B	3.7×10 ⁹	8.8×10 ⁹	-	751, 754	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	0	0	0	
		C	3.3×10 ⁹	1.1×10 ¹⁰	-	746, 748		-	-	0	0	0	
		A _{ref}	1.0×10 ¹⁰	1.9×10 ¹⁰	-	756, 758		-	-	0	0	0	
PS14	<i>Lact. acidophilus</i> , <i>Bif. animalis</i> subsp. <i>lactis</i>	A	5.4×10 ⁸	3.0×10 ⁷	-	341, 345	<i>Lact. crispatus</i>	-	-	0	0	0	NA
		B	8.9×10 ⁸	1.2×10 ⁸	-	339, (343), 340	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	0	0	0	
		C	2.3×10 ⁸	6.5×10 ⁶	-	351, 354	<i>L. reuteri</i>	-	-	0	0	0	
		A _{ref}	3.7×10 ⁸	1.7×10 ⁸	-	347, 348		-	-	0	0	0	

Table 1 (continued)

Product code	Claimed	cfu/dose	City	Actual anaerobic count			Strain code	Identification	Actual aerobic count				NPL#	Identification
				MRS _c	TOS _{MUP}	RCM			KF S _{TTC}	M17 _{lac}	NA	SDA	MK	
PS15	<i>Cl. butyricum</i> and <i>Bifido-bacterium</i>	Not mentioned	A	-	6.4×10 ⁵	6.2×10 ⁵	738, 739	<i>Cl. butyricum</i>	-	-	0	0	0	NA
			B	-	4.2×10 ²	6.3×10 ²	737, 740	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	0	0	0	-
			C	-	-	-	-	-	-	-	-	-	-	-
FPS01	<i>L. rhamnosus</i> GR-1 and <i>L. reuteri</i> RC-14	5 billion viable cells	A _{ref}	-	3.9×10 ⁷	2.1×10 ⁶	734, 732	<i>L. rhamnosus</i>	-	-	0	0	0	NA
			-	1.8×10 ⁸	-	-	998	<i>L. reuteri</i>	-	-	0	0	0	-
			-	-	-	-	1000	-	-	-	0	0	0	-
PS16	<i>Lact. aci-dophilus</i> DDS-1, <i>L. Bulgaricus</i> , <i>Bifido-bacterium infantis</i>	8 billion CFU/Sachet (10g)	A	-	-	-	-	NA	-	-	-	-	-	-
			B	-	-	-	-	-	-	-	-	-	-	<i>B. subtilis</i>
			C	-	-	-	-	-	-	-	-	-	-	<i>Staph. epidermidis</i>
PS17	<i>Lact. aci-dophilus</i> , LA-5; <i>Bifidobacterium</i> , BB12 [®] , <i>Strep. thermophilus</i> , STY-31; <i>Lact. delbrueckii</i> subsp. <i>bulgaricus</i> , LBY-27	>8 billion CFU/sachet (2 g) OR >1 billion CFU/capsule (180mg)	A _{ref}	0	0	-	-	-	-	-	9.9×10 ⁴	0	0	725, 726, 727, 728
			A	2.1×10 ⁹	2.2×10 ⁹	-	992, 990	<i>Lact. crispatus</i>	-	0	0	0	0	-
			B	1.0×10 ⁵	2.3×10 ⁷	-	994	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	0	0	0	0	-
			C	5.9×10 ⁹	7.3×10 ⁹	-	920, 917	-	-	0	0	0	0	-
			A _{ref}	3.3×10 ⁹	2.6×10 ⁹	-	988, 986	-	-	4.4×10 ⁷	4.4×10 ⁷	0	0	901, 902, 903, 904
PS18	<i>Lact. aci-dophilus</i> LA-G80, <i>L. rhamnosus</i> Lr-G14, <i>Bif. lactis</i> BL-G101, <i>Bif. bifidum</i> BB-G90	8 billion CFU/sachet (2g)	A	0	0	-	-	-	-	-	0	-	-	NA
			B	-	-	-	-	-	-	-	-	-	-	-
			C	-	-	-	-	-	-	-	-	-	-	-
			A _{ref}	9.6×10 ⁷	4.9×10 ⁸	-	914, 911	<i>Lact. crispatus</i>	-	-	0	0	0	-
			-	-	-	-	-	<i>Bif. animalis</i> subsp. <i>lactis</i>	-	-	-	-	-	-

Table 1 (continued)

Product code	Claimed	City	Actual anaerobic count			Strain code	Identification	Actual aerobic count				NPL#	Identification
	*Strain	cfu/dose	MRS _c	TOS _{MUP}	RCM			KF S _{TTc}	M17 _{Lac}	NA	SDA	MK	
FPS02	DE111™ (<i>B. subtilis</i>) <i>L. rhamnosus</i> <i>L. casei</i> <i>Bif. longum</i> <i>Lact. acidophilus</i> <i>L. plantarum</i> <i>Bif. breve</i>	5.75 billion active cells per serving	0	0	-	-	NA	-	-	0	0	0	NA

A: Lahore; B: Islamabad; C: Faisalabad; A_{ref}: Lahore (4 °C); A_d: Lahore duplicated sample

PS, probiotic supplements; FPS, foreign probiotic supplements; IF, Infant formula

MRS_c, De Man, Rogosa Sharpe medium supplemented with 0.05% L. cysteine HCl; TOS_{MUP}, transgalactosylated oligosaccharide medium with mupirocin; RCM, Reinforced Clostridial Medium; KF S_{TTc}, Kenner fecal streptococcal agar supplemented with 1% (2, 3, 5-Triphenyl Tetrazolium Chloride); M17_{Lac}, M17 medium supplemented with lactose; NA, nutrient agar; SDA, Sabouraud Dextrose Agar; MK, MacConkey agar

*New nomenclature for the *Lactobacillus reuteri* is *Limosilactobacillus reuteri*; *Lactobacillus fermentum* and *Lactocaseibacillus rhamnosus* is *Lactobacillus rhamnosus*

Strain Typing

Ribotyping Strain level differences were delineated using an automated RiboPrinter microbial characterization system (EI du Pont de Nemours and Co., Wilmington, DE, USA). Following the manufacturer's procedure, the prepared sample was subjected to digestion using the *EcoR*I restriction enzyme, then separated on an agarose gel and subsequently transferred onto a nitrocellulose membrane, followed by hybridization, which yielded a ribotyping pattern. The banding pattern so generated was analyzed using the system's built-in proprietary software.

Multilocus Sequence Typing (MLST) The polymorphism of DNA sequences was studied using a few select housekeeping genes. A PCR was performed to amplify gene fragments from chromosomal DNA of the strains by using oligonucleotides primers as reported elsewhere [24–31] (Supplementary Table 2). The sequences obtained for all isolates were compared for each locus, and allele numbers were assigned to each sequence. Each isolate was defined by an allele profile or sequence type (ST) derived from the combination of numbers corresponding to the alleles at the analyzed locus (Table 2). Sequences that were different even at a single nucleotide site were considered distinct alleles.

In Vitro Probiotic Characterization of Commercial Strains

The potential probiotic assays were performed using new cultures in their logarithmic phases, reconstituted from their cryo-frozen stocks (− 80 °C). The pelleted cells were washed twice with phosphate buffer saline (PBS), re-suspended in the same buffer, and the absorbance was measured at OD₆₀₀ nm using a SpectraMax Plus384 plate reader (Molecular Devices, USA).

Tolerance to GIT Stress

Gastric Acidity Tolerance and Bile Resistance The tolerance to acidic gastric pH (1.5 and 3.0) and high bile (0.1% and 0.3%) (w/v) was determined according to previously described procedures [32] with slight modifications. Briefly, the pH stress resistance assay was performed by exposing the cultures to PBS adjusted to pH 1.5 and 3.0 for 3 h, followed by their recovery in a fresh enriched medium to monitor the changes in overall viability. Percent survival was calculated against their non-stressed or no bile controls.

Lysozyme and Phenol Resistance Resistance to phenol (0.4%) and lysozyme (100 mg/μL) was determined using

Fig. 1 The evolutionary history of bacterial isolates from commercial products based on 16 rRNA sequences was derived using the maximum likelihood method and the Jukes-Cantor model. The tree with the highest logarithmic probability (−918.95) is displayed. The percentage of trees in which the associated taxa are clustered is displayed above the branches. Initial tree(s) for heuristic search were automatically obtained by applying Neighbor-Join and BioNJ algorithms to a matrix of pairwise distances estimated using the Jukes-Cantor model and the topology with superior log likelihood score was selected. The tree is drawn to scale, with branch lengths measured in relation to the number of substitutions per location. This analysis included 69 nucleotide sequences. All positions containing gaps and missing data were eliminated (complete deletion option). There were a total of 231 positions in the final dataset. Evolutionary analyses were performed in MEGA X

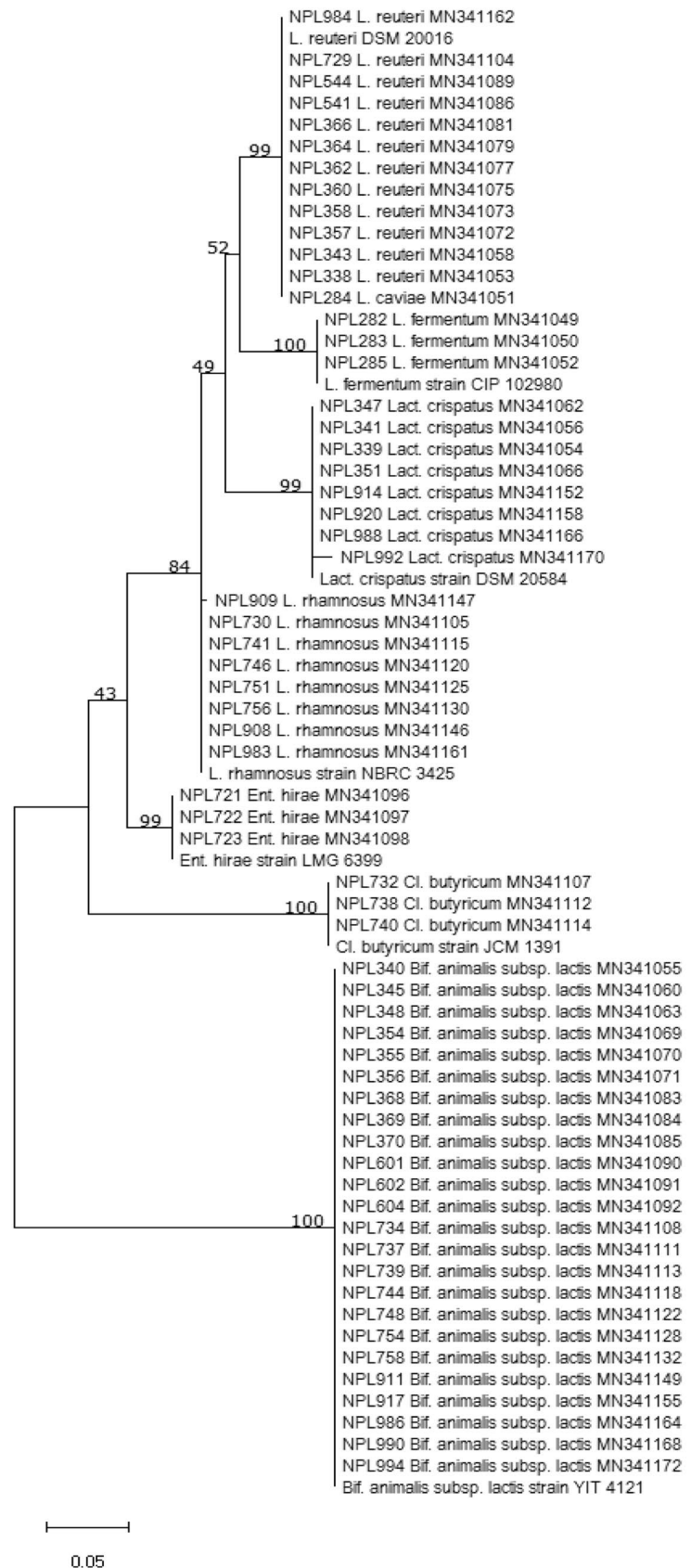


Table 2 Strain typing of supplement isolates via MLST and ribotyping

Species	NPL #	RT	ST	Allele no at locus					Source	Strain claim
<i>L. fermentum</i>				<i>UVrc</i>	<i>pepX</i>	<i>recA</i>	<i>rpoB</i>	<i>dnaA</i>		
	282	1	1	1	1	1	1	1	PS1	<i>L. fermentum</i> (lc40)
	283	1	1	1	1	1	1	1		
	285	1	1	1	1	1	1	1		
<i>L. reuteri</i>				<i>dltA</i>	<i>ddl</i>	<i>Leus</i>	<i>gyB</i>	<i>recA</i>		
	729	1	1	1	1	1	1	1	PS4	<i>L. reuteri</i> DSM17938
	338	1	1	1	1	1	1	1	PS5	<i>L. reuteri protectis</i>
	284	1	1	1	1	1	1	1	PS6	<i>L. protectis</i> (<i>L. reuteri</i>) DSM17938
	357	1	1	1	1	1	1	1		
	366	1	1	1	1	1	1	1		
	364	1	1	1	1	1	1	1		
	544	1	1	1	1	1	1	1		
	362	1	1	1	1	1	1	1	PS7	<i>L. comfortis</i> DSM17938
	358	1	1	1	1	1	1	1		
	360	1	1	1	1	1	1	1		
	541	1	1	1	1	1	1	1		
	984	2	2	1	2	2	2	1	PS12	<i>L. reuteri</i> RC-14
	343	1	1	1	1	1	1	1	PS14	No claim
<i>L. rhamnosus</i>				<i>recA</i>	<i>groEL</i>	<i>rpoC</i>	<i>dnaA</i>	<i>pheS</i>		
	730	1	1	1	1	1	1	1	PS2	<i>L. rhamnosus</i> GG
	908	1	1	1	1	1	1	1		
	909	1	1	1	1	1	1	1		
	983	2	2	1	1	2	1	2	PS12	<i>L. rhamnosus</i> GR-1
	741	3	3	1	1	1	1	3	PS13	<i>L. rhamnosus</i> GG, LGG
	746	4	4	1	1	1	1	4		
	751	3	1	1	1	1	1	1		
	756	3	1	1	1	1	1	1		
<i>Lact. crispatus</i>				<i>FusAB</i>	<i>Leus</i>	<i>gyrB</i>	<i>recA</i>	<i>dltA</i>		
	341	1	1	1	1	1	1	1	PS14	<i>Lact. acidophilus</i> LA-5
	339	1	1	1	1	1	1	1		
	351	1	1	1	1	1	1	1		
	347	1	1	1	1	1	1	1		
	992	2	2	1	1	2	1	1	PS17	<i>Lact. acidophilus</i> LA-5
	920	1	1	1	1	1	1	1		
	988	1	1	1	1	1	1	1		
	914	3	1	1	1	1	1	1	PS18	<i>Lact. acidophilus</i> LA-G80
<i>Ent. hirae</i>				<i>gdh</i>	<i>psts</i>	<i>gkl</i>	<i>xpt</i>	<i>aro</i>		
	721	n.d	1	1	1	1	1	1	PS11	<i>Ent. faecium</i> SF80
	722		1	1	1	1	1	1		
	723		1	1	1	1	1	1		
<i>Cl. butyricum</i>				<i>ddl</i>	<i>groE</i>	<i>Gyr</i>	<i>rpoB</i>	<i>dnaK</i>		
	738	n.d	1	1	1	1	1	1	PS15	<i>Cl. butyricum</i>
	732		2	1	2	1	1	1		
	740		1	1	1	1	1	1		

Table 2 (continued)

Species	NPL #	RT	ST	Allele no at locus					Source	Strain claim
<i>Bif. animalis</i> subsp. <i>lactis</i>				<i>FusAB</i>	<i>rplB</i>	<i>rpoC</i>	<i>dnaJ</i>	<i>HSP60</i>		
	355	n.d	1	1	1	1	1	1	PS8	<i>Bif. animalis</i> subsp. <i>lactis</i> (BB-12)
	356		2	1	1	2	1	1		
	369		3	1	1	3	1	1	PS9	<i>Bifidus</i> B _L BB-12
	601		4	2	1	4	1	1		
	602		5	2	1	1	1	1		
	370		6	1	1	5	1	1		
	604		5	2	1	1	1	1	PS10	<i>Bifidus</i> B _L BB-12
	368		6	1	1	6	1	1		
	744		7	1	1	7	1	1	PS13	<i>Bifidobacterium</i> BB-12
	754		3	1	1	3	1	1		
	748		8	1	1	8	1	1		
	758		9	1	1	9	1	1		
	345		10	1	1	10	1	1	PS14	<i>Bif. animalis</i> subsp. <i>lactis</i> (BB-12)
	340		11	1	1	11	1	1		
	354		3	1	1	3	1	1		
	348		12	1	1	12	1	1		
	739		3	1	1	3	1	1	PS15	<i>Bifidobacterium</i>
	734		13	1	1	13	1	1		
	737		14	1	1	14	1	1		
	990		15	1	1	15	1	1	PS17	<i>Bifidobacterium</i> BB-12
	994		16	1	1	16	1	1		
	917		17	2	1	17	1	1		
	986		18	1	1	18	1	1		
	911		19	2	1	19	1	1	PS18	<i>Bifidobacterium lactis</i> BL-G101

RT, ribotypes; ST, sequence type

The digits represent the different consensus sequence from other; even a change at a single site generates a unique sequence

a reported method [33] with slight modification. The bacterial strains were grown in their respective growth media supplemented with 0.4% phenol or 100 mg/μL of lysozyme, and the OD_{600 nm} was measured at 0 and 24 h and the percent survival rates were calculated.

Adhesion Potential

The adhesion capabilities of strains were measured by determining their cell surface properties such as cell autoaggregation and hydrophobicity following described methods [33].

Autoaggregation For autoaggregation, the bacterial suspension (OD_{600 nm} 0.5 ± 0.05) in sterile PBS was allowed to stand undisturbed for 5 h at room temperature, following which the OD_{600 nm} of the top (1 cm) was measured, and results were recorded as:

$$\text{Autoaggregation\%} = \frac{A_o - A_t}{A_o} \times 100 \quad (1)$$

where A_o and A_t are the absorbance at time 0 and 5 h.

Bacterial Adhesion to Hydrocarbon (BATH) Equal volumes of bacterial suspensions (OD_{600 nm} 0.5 ± 0.05) and xylene were vortexed, and the layers were allowed to separate. The OD_{600 nm} of the aqueous layer before (A_o) and after (A_t) were taken to calculate hydrophobicity as follows:

$$\text{hydrophobicity\%} = \frac{A_o - A_t}{A_o} \times 100 \quad (2)$$

Biofilm Formation Ability The ability of a strain to form biofilms was assessed using the microtiter plate-based crystal violet method, as detailed before [34]. Equal volumes (200 μL) of the resuscitated strains grown to log-phase were added to wells of a polystyrene plate and statically incubated

for 72 h at 37 °C. The wells were then washed with sterile PBS and the biofilms quantified by staining them for 30 min with crystal violet (0.1% w/v). The wells were then washed with distilled water, 3–4 times to get rid of excess stain, and the stain from the adherent cells was extracted with 96% ethanol and its absorbance $A_{570\text{ nm}}$ measured. Sterile culture medium was included as a negative control.

Antimicrobial Activity of Commercial Strains

Commercial strains were evaluated for their antibacterial activities against economically important enteropathogenic microorganisms (Supplementary Table 1) as indicator bacteria. The agar spot method described elsewhere was used [35]. Approximately 10^5 CFU/ μ L of probiotic strains were spot inoculated over the surface of the agar medium surface and allowed to grow for 48 h. The plates were then overlaid with soft TSB agar (0.8% agar w/v) pre-mixed with 10^7 CFU of indicator strain. Post-incubation, the clear zones were measured and recorded.

The plate assay for the H_2O_2 production was done per published report [36] using agar plates mixed with a chromogen TMB (3,3',5,5'-tetramethylbenzidine) and peroxidase enzyme (horseradish peroxidase type II). Following incubation, colonies were observed for blue staining after 30-min exposure to air.

Safety Assessment of Commercial Strains

Antibiotic Resistance and Susceptibility

E-test The sensitivities of the commercial strains to antibiotics were assessed using the E-test method of minimum inhibitory concentrations (MIC) (E-test® bioMérieux, France) as previously described [37]. A set of antibiotics recommended by EFSA were used, and the epidemiological cut-off (ECOFF) values of MICs were determined accordingly [38].

Detection of Resistance Genes Bacterial genomes were examined for the presence of genes involved in resistance to the antibiotic tested using gene-specific primers [39–42] (Supplementary Table 2).

Hemolysis and Gelatinase

Product safety was assessed by testing the strains for their hemolytic potential on blood agar supplemented with 5% human blood and gelatinase activity on nutrient agar supplemented with 5% gelatin. The plates were spotted (~2 μ L) of log-phase cultures of the strains and incubated for 24 h. The blood plates were then observed for zones of hemolysis (alpha (α), beta (β), or gamma (γ) type) and clear halos in case of the gelatin loaded plates [43].

Mucin Utilization and Mucin Degradation

The ability to degrade the mucin was measured as a mucin lysis zone using an agar assay, and the broth assay was performed to check the strains' ability to use mucin following a described method [44]. To check for the ability to utilize mucins, the growth of strains was monitored for 24 h in MRS basal medium (10 g/L proteose peptone, 10 g/L beef extract, 5 g/L yeast extract, 2 g/L ammonium citrate, 5 g/L sodium acetate, 0.1 g/L magnesium sulfate, 0.05 g/L manganese sulfate, and 1 g/L Tween 80, and 2 g/L dipotassium phosphate) reconstituted with either glucose (20 g/L) or mucin (3 g/L). The results were expressed as growth ($\text{OD}_{600\text{ nm}}$) in reconstituted medium minus the residual growth of strains in media having no fermentable carbohydrate source.

Mucin degradation ability was checked by spotting fixed volumes of log-phase culture (~2 μ L) on MRS basal agar medium reconstituted with mucin (3 g/L). MRS agar plates having (20 g/L) served as control. After the plates were incubated for 72 h at 37 °C, their surfaces were flooded with amido black (3 g/L prepared in 3.5 M acetic acid) for 30 min. The plates were then repeatedly washed until clear halos were evident if the mucin was degraded.

Potential to Produce Biogenic Amines

The ability to produce biogenic amines by strains was qualitatively checked on decarboxylase media substituted with 2% L-histidine-monohydrochloride, L-tyrosine disodium salt, or L-ornithine monohydrochloride as described before [45]. The strains were initially cultured in decarboxylating broth having 0.1% w/v of one of the aforementioned amino acid precursors. The activated cultures were spotted (2 μ L) on agar plates of the same decarboxylase media having amino acid precursors and incubated anaerobically for 72 h at 37 °C. A clear halo zone was indicative of tyramine formation, and a purple color showed the presence of histamine or putrescine.

Functional Characterization of Commercial Strains

Antioxidant Activity

The commercial strains were tested for their antioxidative potential by two scavenging assays of hydroxyl radicals and superoxide anions (pyrogallol autoxidation) described elsewhere in detail [46].

Hydroxyl Radical Scavenging Assay Broth cultures of strains in their log phases were harvested by centrifugation and washed using sterile PBS. Equal quantities of 1,10-phenanthroline and FeSO_4 (500 μ L of 6 mmol/L) were added to a sterile test tube and immediately mixed with 1 mL of sterile PBS. It was then topped up with 500 μ L of cell suspension prepared earlier (10^9 CFU/mL) followed by 500 μ L solution

of H_2O_2 (0.1% v/v). The same solution without cells served as a control and the solution containing only 1,10-phenanthroline and FeSO_4 was considered a blank. Sterile distilled water was added to make final volume up to 4 mL in all. The $\text{OD}_{536\text{ nm}}$ was measured, and results were calculated using:

$$\text{Hydroxyl radical scavenging (\%)} = \frac{A_s - A_1}{A_0 - A_1} \times 100 \quad (3)$$

where A_s , A_1 , and A_0 were the absorbances of sample, control, and blank, respectively.

Superoxide Anion Scavenging Assay (Pyrogallol Autooxidation) Almost 50 μL of cell suspension, prepared as above, was diluted in 910 μL of sterile water and immediately mixed with 2 mL Tris–HCl (50 mmol/L, pH 8.2) buffer. Then, 40 μL of pyrogallol solution (10 mmol/L) was added and autooxidation was determined at $\text{OD}_{325\text{ nm}}$ after every 1 min for 10 min, to ascertain the maximum inhibition point of reaction. Deionized water instead of cell suspension acts as control and the results are calculated using the following:

$$\text{Superoxide anion scavenging activity (\%)} = \frac{\Delta A_0 - \Delta A}{\Delta A_0} \times 100 \quad (4)$$

where ΔA_0 and ΔA represent the pyrogallol rate of autooxidation before and after the addition of the sample and water, respectively.

Bile Salt Hydrolysis

The ability of commercial strains to deconjugate the bile salts was examined as previously described [47]. The BSH plates were prepared by supplementing the MRS_c agar with 0.5% (w/v) bile salts (deoxycholate, taurodeoxycholate, tauroglycocholate, and taurocholate) and 0.37 g/L of CaCl_2 . The tested strains were spot plated and the precipitation zone of hydrolyzed salts surrounding the colonies was measured (mm) using a Vernier caliper.

Cholesterol Assimilation

The commercial probiotics were tested to see whether they could assimilate cholesterol using a previously described method [46]. The broth medium was supplemented with cholesterol PEG 100 $\mu\text{g/mL}$, and the bacterial strains were allowed to grow for 48 h. The supernatant (500 μL) was collected and mixed with 33% (w/v) KOH solution (500 μL) and absolute ethanol (1 mL), vortexed, and incubated at 37 °C for 15 min. One milliliter of deionized water and 1.5 mL of hexane were added and left at room temperature for phase separation to occur. The hexane layer was transferred to another tube, and once dry, the o-phthalaldehyde (50 mg/dL) reagent prepared in acetic acid was mixed

properly. H_2SO_4 (250 μL) was added, followed by the incubation for 20 min and the absorbance read at 570 nm. The amount of assimilated cholesterol was calculated using a standard curve with concentrations (0, 20, 40, 60, 80, and 100 $\mu\text{g/mL}$) plotted against absorbance values.

INFOGEST Static In Vitro Simulation of GIT

The survival of the commercial probiotic supplement under harmonized static GIT conditions was investigated using the COST Infogest digestion model [48]. The products were subjected to sequential human physiological oral, gastric, and intestinal digestive parameters. Enzymatic activity, bile salt concentrations, and pH-test adjustment (to measure and record the amounts of HCl/NaOH used for attaining target pH) were made according to the standardized recommended assays. The simulated digestive fluids made were simulated salivary fluid (SSF), simulated gastric fluid (SGF), and simulated intestinal fluid (SIF). All the fluids were prepared in their respective electrolyte stock solutions, enzymes, CaCl_2 , and water. A dose of the test supplement was dissolved in 5 mL of sterile PBS and used for the COST experiment. A supplement suspension was made mixed (1:1) with SSF (pH 7) without amylase due to the absence of starch. After 2 min with SSF, the bolus suspension was mixed (1:1) with SGF (pH 3) and pepsin for 120 min. Finally, the gastric chyme was mixed (1:1) with SIF containing bile (10 mM/L) and pancreatin (100U trypsin activity/mL). The SGF and SIF incubations were carried out at 37 °C under anaerobic conditions. Sample aliquots were taken before and after every digestion treatment, and the bacterial count was performed as (CFU/mL). Survival rates were calculated and recorded for further analysis.

Effect of Non-antibiotic Drugs on Viability of Commercial Strain Under GIT

The extent to which non-antibiotic drugs affect the viability of probiotic strains was investigated using a GIT simulation assay. Commonly used NSAIDs, acetaminophen, and the anti-diabetic drug metformin were used. The drug and the test supplement (one dose each) were mixed to make 5 mL of ingestion suspension, and the GIT simulation assay followed as described above.

Statistical Analysis

The data were expressed as the mean $\pm$ SD of three experiments. Statistical analysis on GIT simulation and effect of non-antibiotic drugs on probiotic survival was performed using Graph Pad Prism software version 5.01 (GraphPad Software, Inc., La Jolla, CA, USA), applying the one-way ANOVA using Tukey's multiple comparison test. A P -value < 0.05 was considered statistically significant.

Results

Sampling

In this study, the supplement products collected from the national market were analyzed, and their labels were checked to match the guidelines of probiotic products labels. No product label showed approval of any local regulatory agency. All the products claimed to have probiotics, except some (PS9, PS10, and PS15), gave a strain level nomenclature. A tendency to use ambiguous non-scientific terms for species such as *L. protectus* and *L. comfortis* was also observed. No specific CFU/dose was mentioned for the infant formula products (PS6, PS7, PS9, and PS10). Only two products, PS4 and PS5, with live culture claims of *Limosilactobacillus reuteri* cultures, were available. No evidence of scientific literature or any clinical studies has been given on the product labels.

Microbial Enumeration

In this study, 11 mono, four dual, and three multi-strain products were tested for their microbiological claimed content, and the results are presented in Table 1. For comparison, two products coded with FPS1 and FPS2 from the international market were also analyzed. On average, 39% of the product quality tested was poor in cell viability and species information. Considerable batch-to-batch variation was observed in five (~27.78%), and only 33.3% of the products were found suitable and as stated on the label. Two of the mono-strain products (PS5 and PS8) corresponded to their CFU and species specification, whereas in product PS2, this was true only when retailed under refrigerated conditions (4 °C). Although the products PS6 and PS7 had no explicit dose information, their CFU dose/gram corresponded to the GRAS detector (i.e., 10⁶/g) (GRN410 2011). Another mono strain product (PS11) containing *Enterococcus faecium* matched its CFU specification, but 16S rRNA sequencing revealed that the microbe carried belonged to *Enterococcus hirae*. Inconsistencies were found in different batches of PS4 with no viable counts in its two replicates (A&C) or low viable counts (~10² CFU/dose) in another under the recommended culture conditions. An average of 2.9 × 10⁵ CFU/dose was found in PS1 that has a claim of having 3 × 10⁹ CFU/dose of *Limosilactobacillus fermentum* Lc40 in it. Only a modest increase in the recovered dose of 2.7 × 10⁷ CFU/dose was seen when the product was sold under refrigerated conditions. The supplement PS3 was free of any active *Lactobacillus acidophilus* cells when processed under standard conditions. PS13 corresponded to its type and dose statement for the products with dual strain claim compared to others, albeit with lower numbers. A *Lactobacillus* species, *Lact. reuteri*, was recovered from one of the

replicates of the product PS14, having no explicit claim. A relatively poorer quality was observed for the multi-strain product since none of its replicates contained all the advertised strains, and only two strains could hardly be recovered from them. No viable bacterial species was revived from the mixed foreign probiotic sample. The capsule formulations of PS17 followed the CFU claim; however, a discrepancy was observed in the powdered (sachet) formulation of the same product. Only one species of *Bifidobacterium animalis* (NPL994) was resuscitated from the multi-strain product PS17, and the refrigerated replicate D was contaminated with *Proteus mirabilis* and *Bacillus nakamurai*. A total of five products from four different brands were contaminated with *Staphylococcus epidermidis*, *Burkholderia singularis*, *Streptomyces asenjonii*, *Bacillus subtilis*, *Bacillus haynesii*, *B. nakamuri*, and *Pr. mirabilis* (Table 1).

Species and Strain Level Identification

The isolated strains were characterized biochemically, and their growth kinetics were assessed (Supplementary Table 3). Gram-positive bacteria were all catalase-negative shaped either as rods, cocci, or y-shaped under bright field microscopy (BX 63 Olympus Japan). Except for two species, i.e., *Lact. acidophilus* and *Ent. faecium*, identified as *Lactobacillus crispatus* and *Ent. hirae*, respectively, the blast results of the partial 16S rRNA sequences corresponded to their marketed identification. The rRNA sequences of 64 commercial strains were mapped to infer their phylogenetic relationships (Fig. 1). The strains were divided into two main clusters, with all *Bif. animalis* strains grouped as one and others in another. The tree confirmed the similar species with their respective type strains clustered together.

Multilocus sequence type (MLST) and ribotyping (RT) were performed in parallel to infer variations of strains of the same species between or within the product (Table 2). It was observed that four out of twelve strains present in all the products surveyed had variation and were non-clonal, including LA-5, LGG, BB12, and the *Clostridium butyricum* strain from PS15. In particular, no strain variations were observed for *L. reuteri* and *Enterococcus* species. Identical strains of *L. reuteri* DSM17938 were found in four products based on their matching sequence types (STs) and ribotypes (RT). Interestingly, the *L. reuteri* strain (NPL343) identified as foreign to PS14 (no claim of its presence in the product) also created RT and ST like the one produced for DSM17938. Intra-product variation was observed in the LGG strains in PS13 as evidenced by two banding patterns derived from RT and an SNP at position 66 from the *pheS* gene, which confers three STs in MLST. Notably, the RT pattern generated for PS13 LGG strains differed from those generated with PS2 with the same strain statement.

The two products (PS14 and PS17) claiming to have seven *Lact. crispatus* strains and *Lact. acidophilus* LA-5 produce a similar typing pattern except for NPL992, which had a different sequence of housekeeping gene *gyrB* in a different ST and a different banding pattern of the same support the case. A variation within product strains was observed for *Clostridium* NPL732 that produces a different sequence type due to the substitution of A/G at position number 814 in *groE*. 19 STs for 24 *Bif. animalis* strains from 8 different products were generated based on 19 alleles for the locus *rpoC* gene and three for *fusAB*.

In Vitro Probiotic and Functional Characterization of Commercial Strains

Probiotic characterization of commercial strains using in vitro methods was carried out, and the data were subjected to a hierarchical cluster analysis (HCA) which divided the strains into two main clusters (Fig. 2). All LAB and *Clostridium* strains were clustered together, and *Bif. animalis* subsp. *lactis*, along with two strains of *Lactocaseibacillus rhamnosus* (NPL751 and NPL998), were separated as cluster II. The commercial strains survived well at acidic pH 3 and 0.1% (w/v) bile. *Lactobacillus* were generally intolerant of a higher acid content (pH 1.5); the best tolerance (29%) was observed with *Lact. crispatus* strains only. All three *Cl. butyricum* and 21% of the *Bif. animalis* strains were robustly tolerant of gastric pH 1.5 with more than 75% survival rates.

In contrast, the *Bif. animalis* strains from infant formulae fare poorly in a strongly acidic environment (pH 1.5). Physiologically relevant bile concentrations (0.3% w/v) were generally hostile (80% decrease in survival rates across all species) except for *Lact. crispatus* and *Clostridium* strains which showed a survival rate of up to 50%. A significant variation in bile resistance (0.3% w/v) between batches was observed for *L. reuteri* strains from PS7, where half of the strains were found to be resistant with 47% survival rates while the rest were sensitive (14% survival). Cluster I LAB strains were all able to withstand lysozyme while *Bif. animalis* strains were invariably intolerant. The ability to withstand phenolics was good in *L. reuteri* RC-14 (NPL984) and *L. fermentum* strains, with an overall survival rate of 88%. No significant disparity in adhesion potential was recorded for strains of *L. rhamnosus*, e.g., strains from product PS2 exhibited low autoaggregation with moderately hydrophobic cell surface (50%) coupled with a modest capacity for biofilm formation.

Similar was the case of *L. rhamnosus* LGG strains from dual strain products PS12 and PS13 that were also modest in their biofilm forming capacity. Interestingly strains of other species from these products, PS12 (*L. reuteri* RC-14) and PS13 (*Bif. animalis* subsp. *lactis*), were also unremarkable in making biofilms, even though *Bif. animalis* strains

from the latter were significantly hydrophobic. Overall, the strains in cluster II showed good overall antioxidative potential, with scavenging rates for the superoxide anion from 45 to 93%. Hydroxyl anion scavenging activity was noticeable in some *Bif. animalis* strains from the multi-strain products, and the *L. reuteri* strains, NPL338 (72%) and NPL729 (49%), which were better than those from products with the indication of live culture in them. Nevertheless, all strains were invariably poor assimilators of free cholesterol, the maximum observed in *Bif. animalis* NPL355 being only 33%. The ability to hydrolyze the bile salt was prominent in all *Bif. animalis* and *Enterococcus* strains and was not found in any other species or genus tested.

None of the commercial strains from any product produced H₂O₂ when tested in vitro (Supplementary Table 4). *Lactobacillus* strains inhibited *Staphylococcus aureus* only weakly, except for *L. fermentum*, which was slightly better, e.g., ≤ 10 mm. *L. rhamnosus* showed moderate level activity against *Escherichia coli* but none against *Bacillus cereus*. All the *Bifidobacterium* strains were weakly antagonistic to *E. coli* and *B. cereus*. Except for *Bifidobacterium* strains of PS17, all were found to inhibit *Staph. aureus* with (NPL601 and NPL602) from infant formula (PS9) being the most antagonistic. *Clostridium* and BB12 strains of PS15 generally showed no or only weak antagonistic activity (≤ 10 mm) against *Pseudomonas aeruginosa*, *Citrobacter freundii*, *Pseudomonas fluorescens*, and *Streptococcus pyogenes*. All *Cl. butyricum* strains from PS15 killed relatively more *Staph. aureus*, *E. coli*, and *B. cereus* than the *Bifidobacterium* strains from the same product.

Safety Assessment of Commercial Strains

The antibiotic sensitivity of commercial strains was determined using both phenotypic and genotypic methods (Table 3). All the strains examined here were resistant to aminoglycosides, except for *Ent. hirae* strains. The prevalence of AMR genes *tetW* and *inuA* in commercial strains was at 91.9% and 59.6% frequency, respectively. Phenotypically, the prevalence for tetracycline resistance was 59.7% contributed by all strains of *Bif. animalis* and *L. reuteri*, while only the strains of *Lact. crispatus* (accounting for 12.9% of total) were resistant to clindamycin. All strains of *L. reuteri* DSM17938 were resistant to more than three classes of antibiotics, which made them multidrug-resistant (MDR) except for NPL729, which was tolerant of only two classes of antibiotics aminoglycosides and tetracycline. *L. rhamnosus* strains were both pheno- and genotypically resistant to aminoglycosides, but the *inuA* gene detected for clindamycin did not lead to increased MIC values.

The presence of *blazA* and *mec* genes in *L. rhamnosus* LGG strains did not render them phenotypically resistant either. The *Lact. crispatus* strains were phenotypically

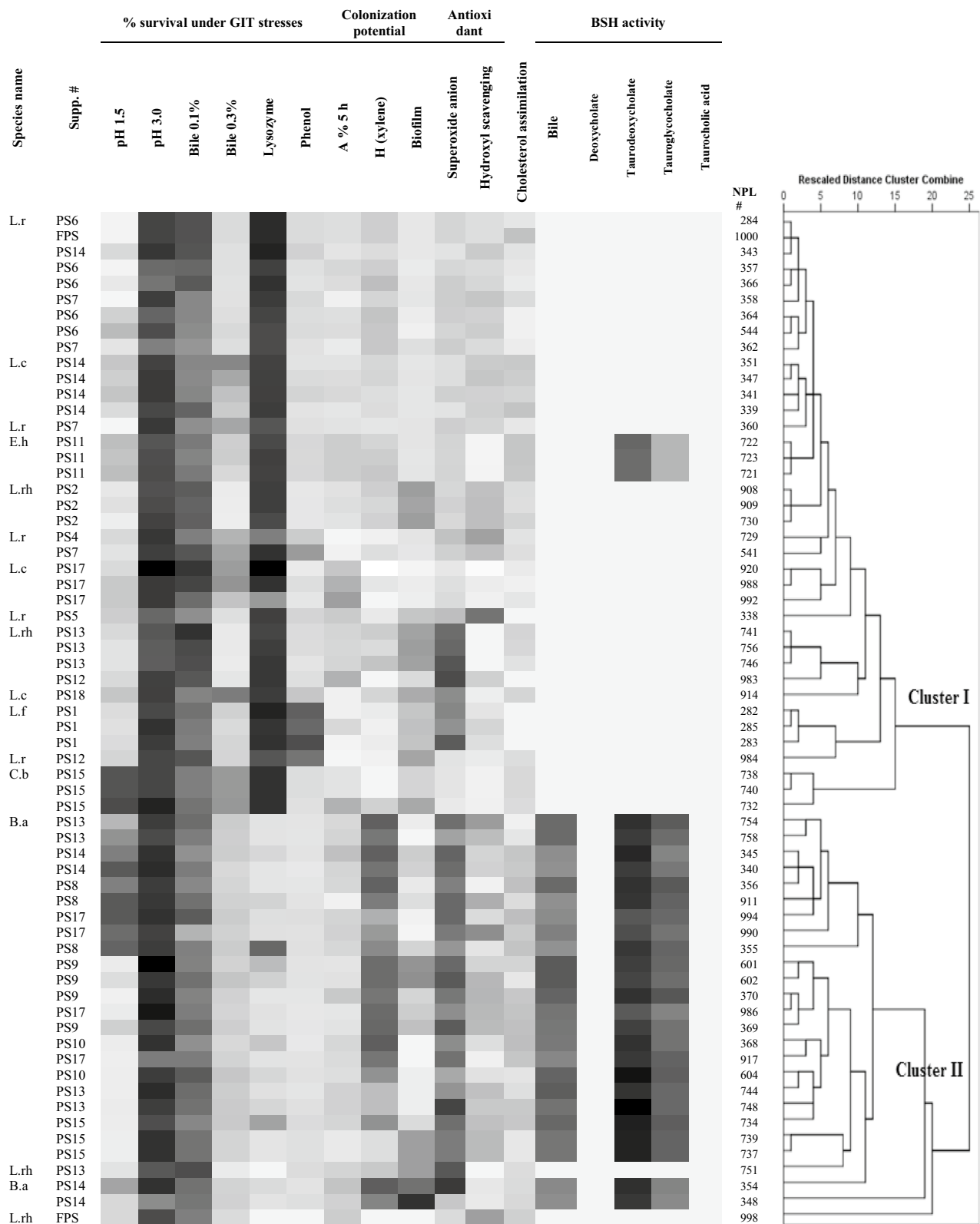


Fig. 2 Heat map, probiotic potential characterization of bacterial isolates from commercial products (white: no activity; black: high activity). Dendrogram resulting from the hierarchical cluster analysis (HCA) using the average linkage to group bacterial ($n=64$) isolates from commercial probiotic products based on their similarity in

survival to various GIT stress elements, colonization and persistence properties, antioxidant activity, cholesterol assimilation and bile salt hydrolase activity. The abbreviations L.r, L.c, L.f, L.rh, E.h, C.b, and B.a denote *L. reuteri*, *Lact. crispatus*, *L. fermentum*, *L. rhamnosus*, *Ent. hirae*, *Cl. butyricum*, and *Bif. animalis* strains respectively

resistant to aminoglycosides and lincosamides. Vancomycin, macrolide, and tetracycline resistance genes were detected in two of three *Cl. butyricum* strains from different lots of PS15, while the third, NPL732, contained *blazA* and *cat* genes. Despite the universal presence of the *tetW* gene in all the commercial strains examined, the occurrence of tetracycline resistance was only found in *L. reuteri* and *Bif. animalis* strains. The *vanA* and *vanE* genes were common in *Bif. animalis* and *L. reuteri* strains, the prevalence rate in the latter being up to 69% but absent in *L. fermentum* and *Enterococcus* strains. *Lact. crispatus* strains (75%) from one product had *van* genes, but none showed any overt signs of vancomycin resistance. Interestingly, the presence of *ermA* genes in our strains also did not make them phenotypically resistant, while some strains of *Bif. animalis* were found to be erythromycin-resistant and carried the *ermB* gene.

The commercial probiotic strains were tested for their hemolytic, gelatinase, mucin degradation and utilization abilities, and biogenic amine production abilities to assess their safety profiles (Supplementary Table 5). All were γ -hemolytic, and none was positive for the ability to degrade gelatin or mucin. Mucin utilization was also not detected in any commercial strains. The qualitative phenotypic detection of histidine, ornithine, and tyrosine decarboxylase activities was carried out in vitro. The strains of *Bif. animalis* strains, apart from those isolated from PS14 and PS8, were positive for histidine decarboxylase. Ornithine decarboxylase activity was detected in two strains of *Cl. butyricum* and all *L. fermentum* strains and relatively rarely in *L. reuteri* and *Bif. animalis* strains. All the commercial LAB strains tested were devoid of tyrosine decarboxylase activity.

INFOGEST Static In Vitro Simulation of GIT

The survival of commercial probiotic strains in the upper gastrointestinal tract was assessed using an in vitro static digestion model. Figure 3A shows the behavior of monostrain products under simulated conditions. Only two live culture claimants, PS5 and PS11, could withstand the physiological GIT conditions. All product strains showed significant tolerance ($P < 0.05$) to salivary digestion, but no viable bacteria were observed after the gastric treatment. Only about 22% of the total products showed post-gastro-intestinal viability and the CFU/mL was significantly reduced statistically ($P < 0.05$). However, for products that are available in both capsule and powder form, the presence of the capsule increases the strain's viability by 1.3 log CFU/dose (PS11). For the PS11, when tested in capsule form, only 10% of the total claimed dose could reach the intestine.

Figure 3B depicts the behavior of mixed strain products. Product PS13 with *L. rhamnosus* and *Bif. animalis* subsp. *lactis* showed varying degrees of tolerance to GIT exposure. The *L. rhamnosus* strain NPL751 survived up to 7.9

logs, while the strain NPL754 *Bif. animalis* was significantly vulnerable, surviving up to 3.6 logs. Another product, PS17, which was available in both sachet and capsules (having claimed doses of 9.9 and 9.0 log CFU respectively for mixed cultures of *Lact. acidophilus* and *Bif. animalis* strains), experienced a significant reduction in viability, both in sachet and capsule form (6.75 and 4.93 log CFU/dose respectively).

Effect of Non-antibiotic Drugs on Viability of Commercial Strain Under GIT

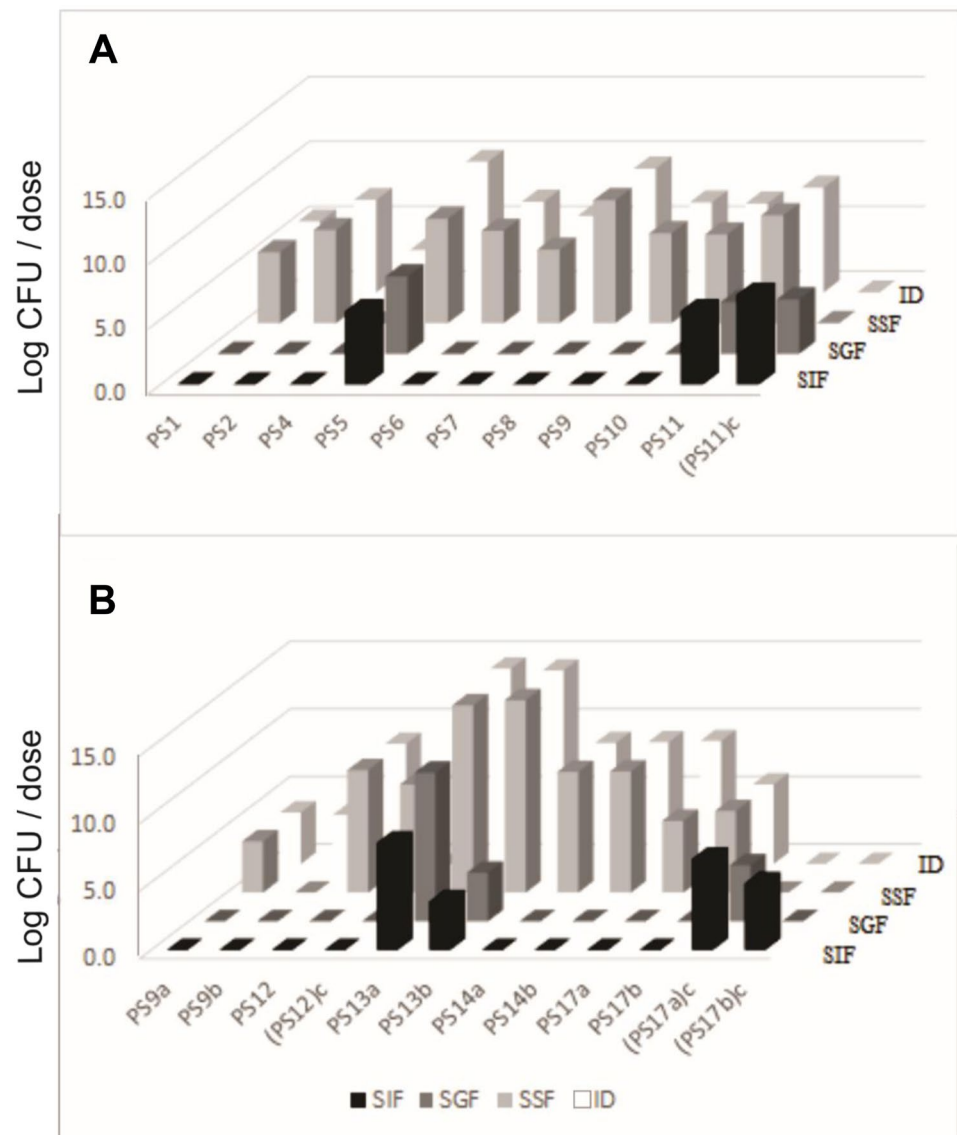
The in vitro digestive model was used to simulate the gastrointestinal behavior of commercial probiotic strains when taken with the most used non-antibiotic drugs (Fig. 4). The effect of drugs on the survival of probiotic strains is highly strain and product specific. Acetaminophen and NSAID have similar effects, i.e., no change in log CFU/dose (no statistically significant changes; $P > 0.05$) observed in the tested *L. reuteri* and *Lact. acidophilus* strains: however, a decrease of 1.1 log CFU/dose was observed in the case of *Ent. hirae* that was also statistically insignificant. However, a significant reduction in CFU/mL ($P = 0.004$) was seen for *L. rhamnosus* strain that was observed to decrease by 3.9 logs with acetaminophen, while NSAID obliterated it. The two strains of *Bif. animalis* strains from PS13 and PS17 responded differently to non-antibiotic drug exposure. Acetaminophen kills *Bif. animalis* from PS13, but the effect statistically less significant compared to the PS17 strain (reduction of 0.2 CFU/dose).

On the other hand, NSAID also caused a slight decrease in the viability of PS17 strains (a 0.2 log CFU/dose decrease) but interestingly stimulated the growth of *Bif. animalis* strain PS13, albeit modestly (0.21 log CFU/dose). The biguanide drug metformin had a markedly statistically significant ($P < 0.05$) stimulatory effect on *L. reuteri* strain DSM17938 (an increase in viability of 67.9%) and less on the strains of *Lact. acidophilus* and *Bif. animalis* (an increase of 0.2–0.4 log CFU/dose). Only the strains of supplement PS13 strains did not respond positively to the drug; the viability of the *L. rhamnosus* strains was almost halved (4.3 logs, ~45%) and for *Bif. animalis* NPL754 strain to zero.

Discussion

The global probiotic market is a multi-billion dollar industry [49] with steady annual growth [50]. Irrespective of local regulatory mechanisms, Asian countries have a prominent role in growing the probiotic market. Increased human awareness and geriatrics are the critical drivers for the lucrative sale of probiotic supplements among Asian countries [7]. Infant dietary probiotic formulas were developed to

Fig. 3 Survival of probiotic bacteria in commercial products while passing through a simulated in vitro digestion model. The terms ID, SSF, SGF, and SIF denote initial dose, simulated salivary fluid, simulated gastric fluid, and simulated intestinal fluid, respectively. Product names with the superscript c at the end imply encapsulated form of product. **A** Single strain. **B** Multistrain. b: *Bif. animalis*; PS9a: *Cl. butyricum*; PS12: *L. reuteri* and *L. rhamnosus*; PS13a: *L. rhamnosus*; PS14a: *Lact. crispatus*; P17a: *Lact. crispatus*

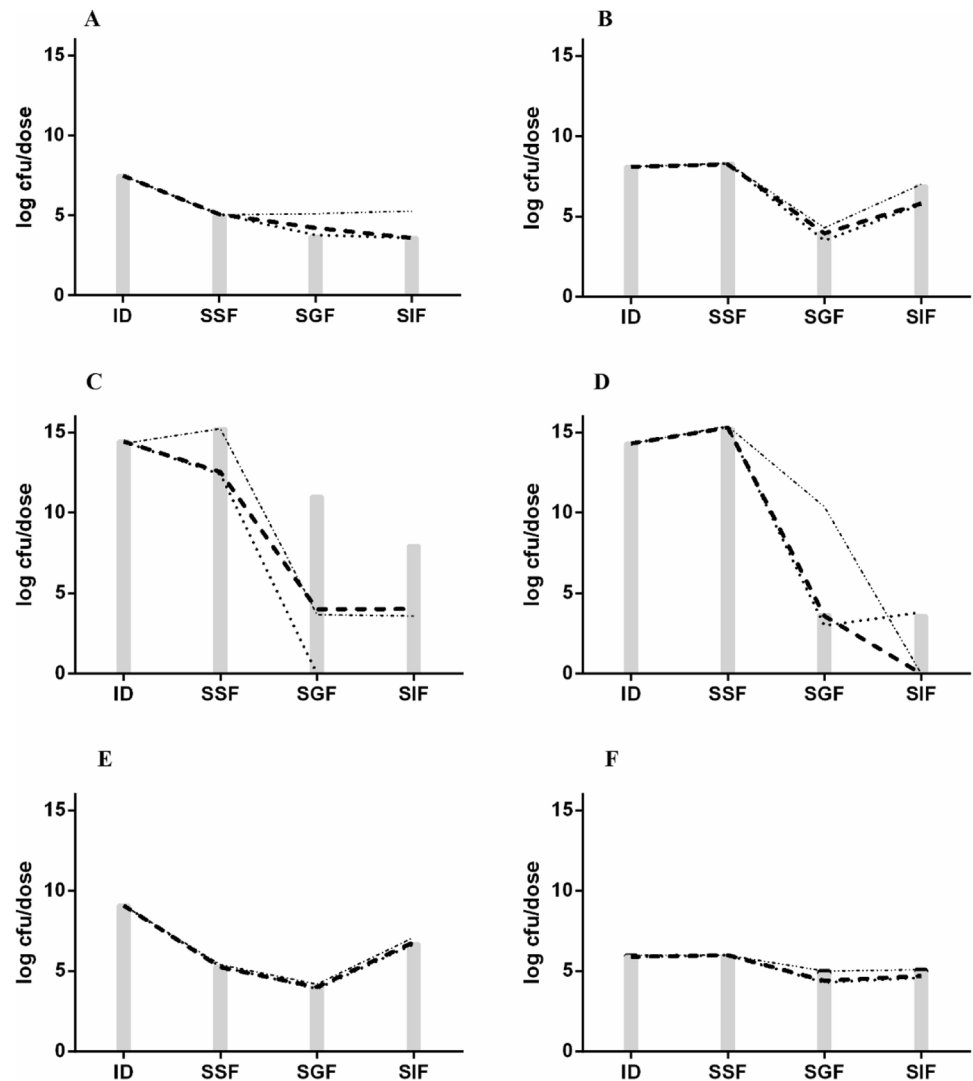


mimic the intestinal flora of breastfed infants [51] and are therefore very popular with parents and health care providers. A previous report evidenced a tragic death of an infant after the probiotic product administration was found later to be contaminated with a fungus and emphasized the need for stringent manufacturing control in production [52]. Therefore, consuming live microorganisms solely based on label claims is not worth the risk [53], and consumers should be wary of this market segment. There is a need for third-party certification to keep consumers and healthcare trust in probiotics [54]. Despite their internationally endorsed definition and increasing commercial interest, probiotics are not given a discrete trade category [55], and thus can be treated in various regulatory infrastructures with varying safety and efficacy requirements [49, 56]. Among South Asian countries, India has put regulatory safeguards to standardize commercial products under the ICMR_DBT guideline [57].

In contrast, the drug regulatory authority in Pakistan refers to the codex standards for regulation of marketed probiotic products [58], without specifying any specific baselines or thresholds for their evaluation. The need for a comprehensive global harmonized framework of probiotic regulation has been debated elsewhere [59]. This study attempts to address the void that arises because of a lack of national-level surveillance and efficacy data on marketed probiotic supplements. Probiotics being dietary supplements are not as rigorously sighted as pharmaceuticals [60]. Numerous studies in different world regions [50, 61–63] on the probiotic product failures form the ground for a much comprehensive study like this.

The prevalent regulatory approach allows producers to make vague and unfounded statements about the health benefits of these products [64]. The lack of information about health claims of any of the locally marketed products

Fig. 4 Effect of various non-antibiotic drugs on the viability of probiotic strains when taken with select supplements. **A** PS5_ *L. reuteri*. **B** PS11_ *Ent. hirae*. **C** PS13_ *L. rhamnosus*. **D** PS13_ *Bif. animalis*. **E** PS17_ *Lact. crispatus*. **F** PS17_ *Bif. animalis*. Vertical gray bars represent growth of test strain log cfu without exposure to any drug, whereas a dashed line, a dotted line, and a stippled line represent exposure to acetaminophen, NSAIDs, and metformin respectively



was telling since for a successful probiotic product, the evidence of health benefits is much more important than the number of strains and highest possible dose [65]. Some structure or functional claims made on the labels were “boost digestive health” or “support the immune system” that were restricted elsewhere [53]. The strain information on the product label allows for better traceability and performance comparison with what is reported in scientific literature. The omission of taxonomical data in more than a third of the marketed products found in this study has a worrisome precedent [15, 66]. Products with complete strain information are generally in line with scientific evidence of probiotics [62, 67–69].

Maintaining strain viability during manufacturing, packaging, and retailing until it reaches the consumer is a pressing challenge. As the probiotic viability decreases with time, the average number must specify the amount at the end of shelf-life. The recommended probiotic dosage must match with the one proven to be most efficacious in a clinical trial

[70]. Accurate and reliable numeracy of probiotic strains is highly dependent on the choice of experimental approach, whose selection must be based on empirical evidence as is the case here [14, 71–74]. This must be accompanied by thorough phenotypic, genotypic characterization involving DNA-based species and strain identification. Most of the brands reported fewer numbers than the recommended and claimed effective dose, and refrigerated storage was not found to significantly improve probiotic viability also reported elsewhere [75]. This less than the optimal probiotic count is likely to reduce the magnitude of health benefit they are expected to deliver since the overcrowded gut typically requires a highly concentrated number of beneficial bacteria to make any impact [16]. The low species and cells revival from the multi-strain products compromised their quality as multi strains are considered more efficacious than the mono strain products (Table 4) [76].

Probiotic benefits are highly specific and only detectable in a particular strain; therefore, the proper documentation

Table 4 Cumulative probiotic score card

Product code	Viability = claimed: 1; < claimed: 0	Claimed sp. Yes: 1; no: 0	Unclaimed sp. No: 1; yes: 0	Strain variation No: 1; yes: 0	Safety safe: 1; unsafe: 0	Survival in GIT Yes: 1; no: 0	Total score Out of 6
PS01	0	1	1	1	1	0	4.00
PS02	0.33	1	1	1	1	0	4.33
PS03	0	0	0	0	0	0	0.00
PS04	0	1	1	1	1	0	4.00
PS05	1	1	1	1	0	0	4.00
PS06	1	1	1	1	0	0	4.00
PS07	0.75	1	1	1	0	0	3.75
PS08	1	1	1	0	1	0	4.00
PS09	1	1	0.5	0	1	0	3.50
PS10	1	1	1	0	0	0	3.00
PS11	1	0	1	1	1	0	4.00
PS12	0.50	1	0.50	1	1	0	4.00
PS13	1	1	1	0	1	0	3.00
PS14	0	0.5	0.25	1	0	0	1.75
PS15	0	1	1	0	1	0	3.00
PS16	0	0	0	0	0	0	0.00
PS17	0.75	0.25	0	0	0	0	1.00
PS18	0	0.25	0.25	1	0	0	1.50

in products is a persistent problem, and it has been demonstrated that products bearing the same labeled strain can have subtly differing 16S sequences and phenotypes [83]. A minimal inter-lot variation in strains is reported elsewhere [62]. The inter-and intra-product LGG strain-level variation in our products is supported from findings reported elsewhere and imputed to a high number of insertion sequences (IS) [84]. Presence of different strains of *Bif. animalis* in probiotic supplements, as found in this study, has not been reported before. The higher the IS element in a genome, the higher the likelihood that a species will have strain variations. Rearrangements or mutations in these elements under various environmental conditions or over the generations [85] can comprise the probiotic functionalities and physiology [86]; therefore, strict strategies must be assured for the probiotic strains to match with its mother strain especially in unstable genome regions.

Environmental bacteria such as *S. asenjonii* reportedly thrive in the extreme conditions of desert soil [87]. *B. singularis* in infant formula has been reported as an opportunistic human pathogen and from environmental sources [88]. Other environmental bugs recovered from the products in this study were *B. subtilis* and *B. nakamurai*, which have a more cosmopolitan distribution [89]. Ubiquitous human skin colonizers, the *Staph. epidermidis*, are also opportunistic pathogens [90], while *Pr. mirabilis*, which is an etiological agent of urinary tract infections (UTIs), has been linked to food poisoning outbreaks in China [91]. Ingestion

of pathogens at intermittent intervals with probiotics can endanger the health of the consumer since the minimum infective dose is highly variable, ranging anywhere from 10 cells in some to 10^3 – 10^8 in others [92]. The pathogenicity of undesirable species in products suggests that the preparatory processes do not conform to safe occupational practices resulting in potentially hazardous products [62, 75], underscoring the need for instituting adequate product safety controls during manufacture [62]; the responsibility for this rests with the probiotic makers [56, 93].

The correct number of probiotic cells in a product does not guarantee its effectiveness. To be effective at the target site in the human gastrointestinal tract, it must survive the harsh physiological conditions that occur therein, i.e., pH, sundry inorganic salts, and enzymes. Tolerance to salivary enzymes (lysozyme), gastric pH, intestinal bile concentrations, and phenol are the top factors in the “step-by-step approach” probiotic selection scheme [9, 94]. The sensitivity of gram-positive bacteria to salivary lysozyme is well documented, with some LAB being the exception [94]. All potentially probiotic strains of *Lactobacillus*, *Clostridium*, and *Enterococcus* were found to be tolerant of higher levels of lysozyme in saliva as reported earlier [95]. However, the sensitivity of *Bif. animalis* strains to lysozyme contradicts previous findings [96]. Following salivary exposure, the next GIT challenge of the probiotic is the gastric transit with its high acid content. The human stomach pH is constantly fluctuating, during fasting and after a meal usually falls within a range of 1 to 4.5, respectively.

Moreover, transit time for food ingestion in the stomach is up to 3 h. Testing for bile concentrations relevant to human gut physiology is imperative for gauging the intestinal stability of prospective probiotic strain. The ability to withstand moderate but not high acid strength by the *Enterococcus* strains here aligns with previous results [97]. Acid tolerance in enterococcal species has been linked to their citrate metabolizing capability enabling them to inhabit various fermented products while at the same time affording them to counter hyper alkaline environments [98]. Robust acid tolerance has been reported from some exotic new strains of *Ent. hireae* [99], unlike the ones examined here, underscoring the need to commercialize newer and more effective enterococcal probiotics in the market. It has been argued that bile tolerance overrides gastric acidity as a criterion of probiotic success because, unlike the former, acid stress can be countered by enveloping probiotics in delivery systems or mixing with food matrices [100]. The *L. rhamnosus* LGG strain has its origin in the human intestinal tract origin with a history of acid and biliary tolerance [101], but this differed from our findings here. It could be because strain-level genome instability during the manufacturing processes can alter the physiology of probiotic strains [86]. A recent finding reported poor survival of *L. rhamnosus* LGG in gastric stress and commented on the role of strain origin versus their response to acidic pH [102]. Interestingly, contrary to what was previously reported for the *Bif. animalis* strain BB12, it grew well in acidic media, but tolerating acidity only up to pH 3.0, not below [102].

Once the probiotic cells have successfully maneuvered the GIT gauntlet and reached the lower reaches of the GIT, they must establish a beachhead before their benefits could manifest. Colonization entails that the strains first adhere to epithelial surfaces, a multifaceted and intricate process depending on a great deal on the physiochemical makeup of the probiotic cell surface. Autoaggregation and hydrophobicity define adhesion potential by ensuring higher cell densities in the gut and improved surface interactions [94]. The marginally better biofilm capacity of *L. rhamnosus* LGG cells relative to their poor hydrophobic and autoaggregation attributes could be ascribed to the pili projecting from their surface [103], which are known from previous works to aid in host colonization [104].

Inhibition zone assay in agar plates is routinely used for detecting antimicrobial production [94]. It is surmised that the beneficial effects on the host primarily stem from the probiotic's ability to displace potential pathogens in the gut [105]. Because of their antagonistic potential, they have been widely used as prophylactic agents against intestinal infections, assuming an increasingly important role as a therapeutic alternative in the face of rising antimicrobial resistance in pathogen and antibiotic ineffectiveness [106].

Regardless of QPS or GRAS awards, strains of the genus *Lactobacillus* and *Bifidobacterium* should be tested for the absence of undesirable properties such as antibiotic resistance, hemolysin, or gelatinase production, and metabolic activity [70]. The safety concerns closely related to probiotics are harmful metabolic activities, the production of toxins, and the potential for pathogenicity [107]. Since the microbes generally represent a significant reservoir of antibiotic resistance [108], those that are to be ingested should be tested for resistance as the gut teems with microbes of all sorts, which magnifies the risk of spread of antimicrobial resistance manifold [109]. Lactic acid bacteria and bifidobacteria are well-known reservoirs of AGRs for clinically relevant antibiotics; therefore, deliberate addition to foods is a potential hazard [38]. The *Lactobacillus* susceptibility to chloramphenicol and erythromycin [42] is endorsed by our findings. Aminoglycoside resistance is typically also present in *Lactobacillus* [110], as seen here except for *L. fermentum* strains. Vancomycin resistance is widespread in *Lactobacillus* spp. and is considered natural [111], but strains of homofermentative Lactobacilli can be sensitive to it [38, 110] congruent to ours in the case of *Lact. crispatus*. The *L. reuteri* DSM17938 strain is derived after removing two potentially transferable antibiotic-resistant plasmids from ATCC 55,730 [112]. The resistance to β -lactamases, tetracycline, and genes of lincosamides resistance in our strain is like that of the parent strain (ATCC 55,730) and can transfer this resistance to the intestinal microbiota. ARGs (antimicrobial resistance genes) for tetracycline are often detected in probiotic genera [108, 110]. Resistance to tetracycline, clindamycin, and chloramphenicol is all conjugatively transferable [111]. The ingestion of such resistant bacteria can lead to broader dissemination of ARGs and reduce the efficiency of antibiotherapies [113]. In probiotic strains, the absence of mobile resistance genes must be ensured before being used in production [53]. With the GRAS status, no further approvals are usually required for use, eliminating the need for strict regulations and thus monitoring the quality of the product [114].

Predicting the performance of a probiotic candidate in the human GIT is key to its eventual success. Recently, a consensus-based in vitro set of conditions has been proposed that simulate in vivo digestive events in humans [48, 115], which was also followed in this study. Genetically identical strains derived from different products responded differently to the simulated GIT digestion cues. The superior survival of the BB-12 strain in GIT is a contentious claim, with reports both arguing for and against it [102, 116]. It is noteworthy that some strains could survive the GIT onslaught but in far fewer numbers than stated on the label or required for any substantial effect. In the probiotic context, the concept of adequate amount does not refer to a specific number, but various dose-range studies recommend a daily dose between

10^8 and 10^{11} CFU. Therefore, depending on the procedure adopted, an effective dose for a specific probiotic product must be documented in a clinical trial and guaranteed up to the shelf-life [74]. Because of the general susceptibility of the probiotic strains to GIT stresses, ensuring sufficient viable numbers at their site of action in human GIT has always been a problem, and several strategies have been developed to circumvent it [117–119]. Our studies showed encapsulation to help increase the strain's survival in the harsh GIT milieu as also described elsewhere [120], but the viability numbers were still insufficient as the right dosage is considered vital for any ameliorative effect [74, 120]. The infant formulas tested did not demonstrate viable content after passage through simulated GIT. However, a clinical study claiming a detectable amount of probiotic recovery has been published elsewhere [121, 122]. Infant food probiotics are encased in protective capsules, and the lack of information on probiotic stability in powdered milk makes them [123] vulnerable to the harsh conditions of GIT. Therefore, more attention must be paid to the production protocols to increase viable cells reaching their target destination.

The adverse effect of non-antibiotic drugs on the intestinal microbiota composition is known [124], and it has been proven that they have antibiotic-like side effects in humans [125]. Taking these drugs is aimed at affecting human cells. However, they also impact the microbiome's composition [125]; sometimes, these compositional shifts favor a particular group or even several microbial taxa [126]. It is believed that metformin's blood sugar-lowering effects are due to its modulation of the gut microbiota makeup [127]. Not many studies have looked at how ingesting non-antibiotic drugs with probiotics can affect them. The loss or decrease in probiotic viability with NSAIDs is consistent with a previously reported decrease of $6.75 \log \text{ CFU/mL}$ [128]. Metformin's beneficial effect on probiotic strains is not a new phenomenon, as the drug combined with probiotics has already been a recommended treatment for colorectal cancer (CRC) and diabetes [129].

Conclusion

Worldwide, the use of probiotics and their products is increasing day by day. Given the importance of probiotics and consumer interest in over-the-counter nutraceuticals, effective post-market surveillance and quality checks are needed to maintain consumer interest in products and keep probiotic science on the path to health. The regulatory authorities need to develop a more in-depth functional basis for effective market surveillance of probiotics.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s12602-022-09931-7>.

Author Contribution Ghazal Aziz: investigation, formal analysis, writing original draft. Arsalan Zaidi: conceptualization, methodology, resources, writing—review and editing, visualization, supervision, project administration, funding acquisition. Muhammad Tariq: validation, resources. Contributions of Ghazal Aziz and Arsalan Zaidi are equal.

Funding This work was supported with a Ministry of Planning, Development & Special Initiatives, (Government of Pakistan) PSDP grant 'Development of a National Probiotic Lab at NIBGE'. The corresponding author being its Project Director.

Data Availability All data generated or analyzed during this study are included in this published article (and its supplementary information files).

Code Availability Not applicable.

Declarations

Consent to Participate Not applicable.

Consent to Publish Not applicable.

Conflict of Interest The authors declare no competing interests.

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